

Lab 3

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Architecture Selection and Justification:

Architecture Selection:

We chose Layered Architecture for the Vaccination Cohort & Dose Scheduling System.

Architectural Choice:

The Layered Architecture divides the system into three main layers: Presentation, Business, and Data. The Presentation Layer handles interaction with citizens, vaccination officers, and certificate verifiers. The Business Layer manages registration, slot booking, vaccination status, dose-related operations, and certificate generation. The Data Layer stores and retrieves citizen, booking, vaccination, and certificate information.

Reason 1 – Clear Separation of Responsibilities:

The vaccination system has different types of operations such as user interaction, vaccination-related processing, and data storage. Separating these responsibilities into different layers makes the system easier to understand, maintain, and modify without affecting unrelated parts of the system.

Reason 2 – Suitable for the System Complexity:

The system mainly performs straightforward functions such as citizen registration, vaccination slot booking, vaccination tracking, and certificate generation. Therefore, Layered Architecture is suitable because it provides the required structure without the additional operational complexity of deploying and managing multiple independent microservices.

Security Advantage:

The Business Layer can control access to sensitive vaccination operations. Authentication and authorization can be handled before allowing vaccination officers to view or modify vaccination records, helping prevent unauthorized access to sensitive citizen and vaccination information.

Performance Benefit:

The separation of the user interface, business processing, and database operations allows each layer to handle its specific responsibilities efficiently. This helps the system process booking and vaccination-status requests efficiently and supports the requirement of providing booking confirmation within 2 seconds.